

vival of the fittest [Kan95]. A short discussion on the origins of particle filtering can be found in [Iba01]. Reference books on the particle filter include [Dou01, Ris04].

Particle filters had their origin in Nicolas Metropolis's work in 1949 [Met49], in which he proposed studying systems by investigating the properties of sets of particles rather than the properties of individual particles. He used the analogy of the card game of solitaire. What is the probability of success in a game of solitaire? The probability may be impossible to compute analytically (because of all of the possible permutations of play). But if a person plays several hundred games and succeeds in a certain proportion of those games, then the probability of success can be approximated on that basis:

$$\Pr(\text{Success}) \approx \frac{\text{Number of successes}}{\text{Number of trials}} \quad (15.1)$$

This simple idea harkens back to the definition of probability in Section 2.1. Given the recent invention of the electronic computer at the time, Metropolis's work was certainly ahead of its time. Now that fast, parallel computers are available, his work is beginning to see its fruition in the methods described in this chapter.

As discussed in Chapter 13, the extended Kalman filter (EKF) is the most widely applied state estimation algorithm for nonlinear systems. However, the EKF can be difficult to tune and often gives unreliable estimates if the system nonlinearities are severe. This is because the EKF relies on linearization to propagate the mean and covariance of the state. Chapter 14 discussed the unscented Kalman filter and showed how it reduces linearization errors. We saw that the UKF can provide significant improvements in estimation accuracy over the EKF. However, the UKF is still only an approximate nonlinear estimator. The EKF estimates the mean of a nonlinear system with first-order accuracy, and the UKF improves on this by providing an estimate with higher-order accuracy. However, this simply defers the inevitable divergence that will occur when the system or measurement nonlinearities become too severe.

This chapter presents the particle filter, which is a completely nonlinear state estimator. Of course, there is no free lunch [Ho02]. The price that must be paid for the high performance of the particle filter is an increased level of computational effort. There may be problems for which the improved performance of the particle filter is worth the increased computational effort. There may be other applications for which the improved performance is not worth the extra computational effort. These trade-offs are problem dependent and must be investigated on an individual basis.

The particle filter is a probability-based estimator. Therefore, in Section 15.1, we will discuss the Bayesian approach to state estimation, which will provide a foundation for the derivation of the particle filter. In Section 15.2, we will derive the particle filter. In Section 15.3, we will explore some implementation issues and methods for improving the performance of the particle filter.

15.1 BAYESIAN STATE ESTIMATION

In this section, we will briefly discuss the Bayesian approach to state estimation. This is based on Bayes' Rule, which is discussed in Chapter 2. This section is based on the presentation given in [Gor93], which is similar to many other books and papers on the subject of Bayesian estimation [Dou01, Ris04].

Suppose we have a nonlinear system described by the equations

$$\begin{aligned}x_{k+1} &= f_k(x_k, w_k) \\ y_k &= h_k(x_k, v_k)\end{aligned}\tag{15.2}$$

where k is the time index, x_k is the state, w_k is the process noise, y_k is the measurement, and v_k is the measurement noise. The functions $f_k(\cdot)$ and $h_k(\cdot)$ are time-varying nonlinear system and measurement equations. The noise sequences $\{w_k\}$ and $\{v_k\}$ are assumed to be independent and white with known pdf's. The goal of a Bayesian estimator is to approximate the conditional pdf of x_k based on measurements $y_1, y_2, \dots, y_k$. This conditional pdf is denoted as

$$p(x_k|Y_k) = \text{pdf of } x_k \text{ conditioned on measurements } y_1, y_2, \dots, y_k \tag{15.3}$$

The first measurement is obtained at $k = 1$, so the initial condition of the estimator is the pdf of x_0 , which can be written as

$$p(x_0) = p(x_0|Y_0) \tag{15.4}$$

since Y_0 is defined as the set of no measurements. Once we compute $p(x_k|Y_k)$ then we can estimate x_k in whatever way we think is most appropriate, depending on the problem. The conditional pdf $p(x_k|Y_k)$ may be multimodal, in which case we may not want to use the mean of x_k as our estimate. For example, suppose that the conditional pdf is computed as shown in Figure 15.1. In this case, the mean of x is 0, but there is zero probability that x is equal to 0, so we may not want to use 0 as our estimate of x . Instead we might want to use fuzzy logic and say that $\hat{x} = \pm 2$, each with a level of membership equal to 0.5 [Lew97].

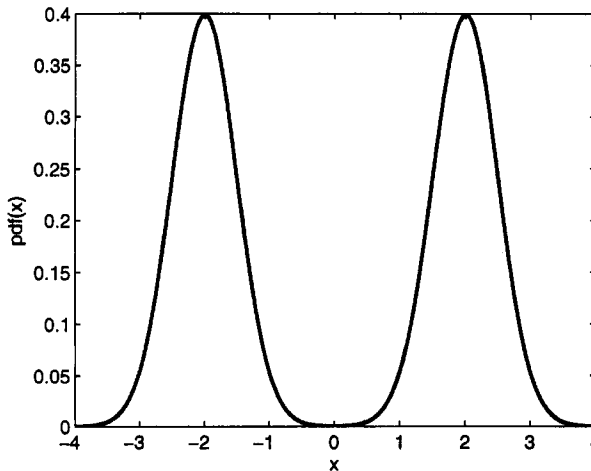


Figure 15.1 An example of a multimodal probability density function. What single number should be used as an estimate of x ?

Our goal is to find a recursive way to compute the conditional pdf $p(x_k|Y_k)$. Before we find this conditional pdf, we will find the conditional pdf $p(x_k|Y_{k-1})$. This

is the pdf of x_k given all measurements *prior to* time k . We can use Equations (2.17) and (2.51) to write this pdf as

$$\begin{aligned} p(x_k|Y_{k-1}) &= \int p[(x_k, x_{k-1})|Y_{k-1}] dx_{k-1} \\ &= \int p[x_k|(x_{k-1}, Y_{k-1})]p(x_{k-1}|Y_{k-1}) dx_{k-1} \end{aligned} \quad (15.5)$$

But notice from our system description in Equation (15.2) that x_k is entirely determined by x_{k-1} and w_{k-1} ; therefore $p[x_k|(x_{k-1}, Y_{k-1})] = p(x_k|x_{k-1})$ and we see that

$$p(x_k|Y_{k-1}) = \int p(x_k|x_{k-1})p(x_{k-1}|Y_{k-1}) dx_{k-1} \quad (15.6)$$

The second pdf on the right side of the above equation is not available yet, but it is available at the initial time [see Equation (15.4)]. Later in this section we will see how to compute it recursively. The first pdf on the right side of the above equation is available. The pdf $p(x_k|x_{k-1})$ is simply the pdf of the state at time k given a specific state at time $(k-1)$. We know this pdf because we know the system equation $f_k(\cdot)$ and we know the pdf of the noise w_k (see Section 2.3). For example, suppose that the system equation is given as $x_{k+1} = x_k + w_k$ and suppose that $x_{k-1} = 1$ and w_{k-1} is uniformly distributed on $[-1, 1]$. Then the pdf $p(x_k|x_{k-1})$ is uniformly distributed on $[0, 2]$.

Now consider the *a posteriori* conditional pdf of x_k . We can again use Equations (2.17) and (2.51) to write this pdf as

$$\begin{aligned} p(x_k|Y_k) &= \frac{p(Y_k|x_k)}{p(Y_k)}p(x_k) \\ &= \frac{p(y_k, Y_{k-1}|x_k)}{p(y_k, Y_{k-1})} \underbrace{\frac{p(x_k|Y_{k-1})p(Y_{k-1})}{p(Y_{k-1}|x_k)}}_{p(x_k)} \\ &= \frac{p(x_k, y_k, Y_{k-1})}{p(x_k)p(y_k, Y_{k-1})} \frac{p(x_k, Y_{k-1})p(Y_{k-1})}{p(Y_{k-1})p(Y_{k-1}|x_k)} \end{aligned} \quad (15.7)$$

We can multiply both the numerator and denominator of this equation by $p(x_k, y_k)$ to obtain

$$p(x_k|Y_k) = \frac{p(x_k, y_k, Y_{k-1})p(x_k, Y_{k-1})p(Y_{k-1})}{p(x_k)p(y_k, Y_{k-1})p(Y_{k-1})p(Y_{k-1}|x_k)} \frac{p(x_k, y_k)}{p(x_k, y_k)} \quad (15.8)$$

Now we use the ratios of various joint pdfs to marginal pdfs in the above equation to obtain conditional pdfs. This gives

$$p(x_k|Y_k) = \frac{p[Y_{k-1}|(x_k, y_k)]p(y_k|x_k)p(x_k|Y_{k-1})}{p(y_k|Y_{k-1})p(Y_{k-1}|x_k)} \quad (15.9)$$

Note that y_k is a function of x_k , so $p[Y_{k-1}|(x_k, y_k)] = p(Y_{k-1}|x_k)$. These two terms cancel in the above equation and we obtain

$$p(x_k|Y_k) = \frac{p(y_k|x_k)p(x_k|Y_{k-1})}{p(y_k|Y_{k-1})} \quad (15.10)$$

All of the pdf's on the right side of the above equation are available. The pdf $p(y_k|x_k)$ is available from our knowledge of the measurement equation $h_k(\cdot)$ and our knowledge of the pdf of the measurement noise v_k . The pdf $p(x_k|Y_{k-1})$ is available from Equation (15.6). Finally, the pdf $p(y_k|Y_{k-1})$ is obtained (in the same way that Equation (15.5) was obtained) as follows:

$$\begin{aligned} p(y_k|Y_{k-1}) &= \int p[(y_k, x_k)|Y_{k-1}] dx_k \\ &= \int p[y_k|(x_k, Y_{k-1})]p(x_k|Y_{k-1}) dx_k \end{aligned} \quad (15.11)$$

But y_k is completely determined by x_k and v_k , so $p[y_k|(x_k, Y_{k-1})] = p(y_k|x_k)$ and

$$p(y_k|Y_{k-1}) = \int p(y_k|x_k)p(x_k|Y_{k-1}) dx_k \quad (15.12)$$

Both of the pdf's on the right side of the above equation are available as discussed above. $p(y_k|x_k)$ is available from our knowledge of the measurement equation $h(\cdot)$ and the pdf of v_k , and $p(x_k|Y_{k-1})$ is available from Equation (15.6).

Summarizing the development of this section, the recursive equations of the Bayesian state estimation filter can be summarized as follows.

The recursive Bayesian state estimator

1. The system and measurement equations are given as follows:

$$\begin{aligned} x_{k+1} &= f_k(x_k, w_k) \\ y_k &= h_k(x_k, v_k) \end{aligned} \quad (15.13)$$

where $\{w_k\}$ and $\{v_k\}$ are independent white noise processes with known pdf's.

2. Assuming that the pdf of the initial state $p(x_0)$ is known, initialize the estimator as follows:

$$p(x_0|Y_0) = p(x_0) \quad (15.14)$$

3. For $k = 1, 2, \dots$, perform the following.

- (a) The *a priori* pdf is obtained from Equation (15.6).

$$p(x_k|Y_{k-1}) = \int p(x_k|x_{k-1})p(x_{k-1}|Y_{k-1}) dx_{k-1} \quad (15.15)$$

- (b) The *a posteriori* pdf is obtained from Equations (15.10) and (15.12).

$$p(x_k|Y_k) = \frac{p(y_k|x_k)p(x_k|Y_{k-1})}{\int p(y_k|x_k)p(x_k|Y_{k-1}) dx_k} \quad (15.16)$$

Analytical solutions to these equations are available only for a few special cases. In particular, if $f(\cdot)$ and $h(\cdot)$ are linear, and x_0 , $\{w_k\}$, and $\{v_k\}$ are additive, independent, and Gaussian, then the solution is the Kalman filter discussed in Chapter 5. This way of obtaining the Kalman filter is more complicated than the least squares approach that we used in Chapter 5. The Bayesian derivation of the